

FOURTH SEMESTER (CBCSS-UG) DEGREE EXAMINATION, APRIL 2022**Mathematics****MEC 4C 04—MATHEMATICAL ECONOMICS****(2019 Admission onwards)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 4C 04--MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. Regression analysis is concerned with the study of the dependence of :
 - (A) Explanatory variables on one or more dependent variables.
 - (B) Dependent variable on one or more explanatory variables.
 - (C) Both explanatory and dependent variables on other known variables.
 - (D) Two known variables.
2. A statistical relationship in itself :
 - (A) Can help establish causation.
 - (B) Can help establish direction of causation.
 - (C) Cannot logically imply causation.
 - (D) Always shows correlation.
3. The dependent variable in regression analysis is assumed to be :
 - (A) Non-stochastic.
 - (B) Constant.
 - (C) Stochastic.
 - (D) Known values.
4. Firm data collected for top 10 companies classified based on profitability for 10 years is an example of :
 - (A) Cross-sectional data.
 - (B) Time series data.
 - (C) Pooled data.
 - (D) Panel data.
5. In $Y_i = \beta_1 + \beta_2 X_i + u_i$, u_i can take values that are :
 - (A) Only positive.
 - (B) Only negative.
 - (C) Only zero.
 - (D) Positive, negative or zero.
6. For a regression line that parses through the conditional means of Y , $E(Y|X_i)$ is :
 - (A) Always a positive value.
 - (B) Always a negative value.
 - (C) Always zero.
 - (D) Any of the above.

7. In the simple linear regression model, the regression slope :
- (A) Indicates by how many percent Y increases given a one percent increase in X.
 - (B) When multiplied by the explanatory variable will give you the predicted Y ?
 - (C) Indicates by how many units Y increases, given a one unit increase in X.
 - (D) Represents the elasticity of Y on X.
8. The statement that-There can be more than one SRF representing a population regression function is :
- (A) Always true.
 - (B) Always false.
 - (C) Sometimes true, sometimes false.
 - (D) Nonsense statement.
9. The least square estimators are :
- (A) Period estimators.
 - (B) Point estimators.
 - (C) Population estimators.
 - (D) Popular estimators.
10. Under normality assumption of u_i , the OLS estimator are :
- (A) Minimum variance unbiased.
 - (B) Consistent.
 - (C) $\hat{\beta}_1$ is normally distributed.
 - (D) All the above.
11. Rejecting a true hypothesis results in this type of error :
- (A) Type I error.
 - (B) Type II error.
 - (C) Structural error.
 - (D) Hypothesis error.
12. In confidence interval estimation, $\alpha = 5\%$, this means that this interval includes the true β with probability of :
- (A) 5%.
 - (B) 50%.
 - (C) 95%.
 - (D) 45%.
13. Given that the test Statistics under test-of-significance approach lies in the critical region, the null hypothesis is :
- (A) Not rejected.
 - (B) Rejected.
 - (C) Rejection depends on α value.
 - (D) Rejection depends on t value.

Turn over

14. A researcher estimates the unknown demand curve $Q = \alpha + \beta P + \epsilon$ and fits the regression line $Q = \alpha - bP$ using OLS technique. The following information is given to you : Sample size = 10, TSS = 80 : ESS = 44. Which of the following is correct ?
- (A) RSS = 5 b. C. d.
 - (B) Given that $a = 4$ and $b = 2$, demand is price unitary elastic at $P = 1.5$.
 - (C) All coefficients are statistically significant at 5% level of significance.
 - (D) $R^2 = 0.25$.
15. The Jarque-Bera test is a :
- (A) Model mis-specification test.
 - (B) Residual normality test.
 - (C) Test of unbiasedness of estimators .
 - (D) Test of goodness of fit for the model.

**FOURTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2022**

Mathematics

MEC 4C 04—MATHEMATICAL ECONOMICS

(2019 Admission onwards)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. What are the different types of data ?
2. Distinguish between regression and correlation ?
3. Define a random or stochastic variable with an example.
4. Discuss the concept of Sample Regression Function (SRF) with specification.
5. What are the properties of R^2 .
6. Distinguish between theoretical Econometrics and Applied Econometrics.
7. What is Homoscedasticity ?
8. What do you mean by multi-collinearity ?
9. What is level of significance ?
10. Briefly state the principle of least squares.
11. Explain semi-log models.
12. What is a standardized variable ?

(8 × 3 = 24 marks)

Turn over

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Explain the difference between statistical and deterministic relationship.
14. What is Gaus-Markov Theorem ?
15. Derive R^2 .
16. Explain the properties of OLS Estimators under the Normality Assumption for U_i .
17. Explain how regression is compared ANOVA.
18. Derive the relationship between R^2 and $\bar{R}^2$.
19. What are the main guidelines for the choice of a functional form of regression ?

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. Explain the traditional methodology of Econometrics to proceed the analysis of an economic problem.
21. Explain the statistical properties of OLS estimators ?

(1 × 11 = 11 marks)